

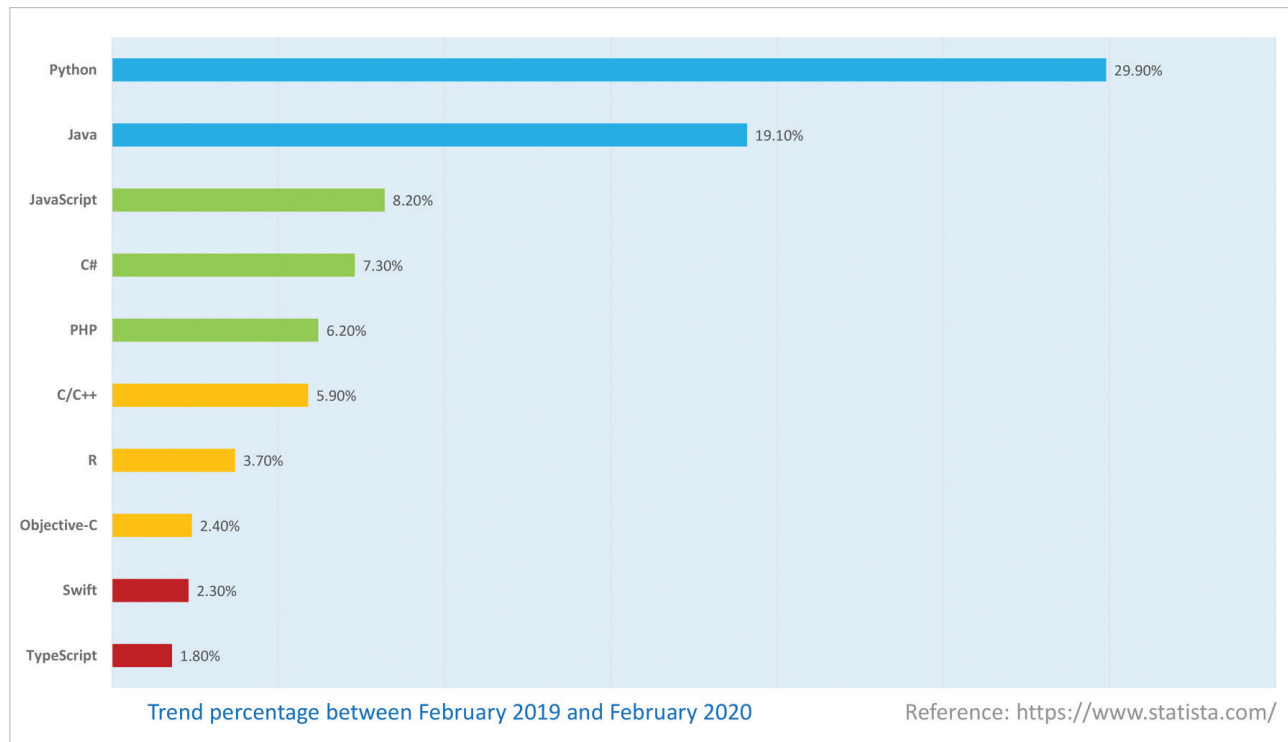


Figure 3: Trend percentages of programming languages

The Tesla Autopilot (advanced driver-assistance system) used in Tesla cars was built using PyTorch.

- KNIME or the Konstanz Information Miner is free software that can do data analysis and reporting using ML and data mining. It integrates various components for agent learning and data mining through its modular data pipelining ‘Lego of Analytics’ concept. It provides a graphical user interface (GUI) and Java database connectivity (JDBC) features for blending various data sources for modelling, data analysis and visualisation without, or with only minimal, programming. It has been used in areas like pharmaceutical research, CRM customer data analysis, business intelligence, text mining and financial data analysis.
- Keras provides a Python interface for artificial neural networks and is well known for its modularity, speed, and ease of use. It supports backends like TensorFlow,

Microsoft Cognitive Toolkit, Theano, and PlaidML. It’s designed to enable fast experimentation with deep neural networks and is user friendly too.

Python for ML

Many types of programming languages like Python, C/C++, Java/JavaScript and R are used for ML but Python is the most widely used because of its simplicity and features. It was created in the late 1980s and was first released in 1991, by Guido van Rossum, as a successor to the ABC programming language.

According to one survey done by Statista (a German company specialising in market and consumer data), Python is the most popular programming language in the world. Unlike any other language, Python gives the options to build ML programs using its robust library and cross-compilation ability. Moreover, the syntax used in Python is simpler than that of C/C++ or Java.

For example, if a program is written to print the line ‘Hello World’ using C /

C++, Java and Python, then it will be done as follows:

```
// Print "Hello World" using C++
Program
#include <iostream>
int main() {
    std::cout << "Hello World!";
    return 0;
}
```

```
// Print "Hello World" using Java
class HelloWorld {
    public static void main(String[]
args) {
        System.out.println("Hello,
World!");
    }
}
```

```
# Print "Hello World" using Python
print('Hello, world!')
```

As you can see, to print ‘Hello World’ using C/C++ and Java, many lines of code are needed, whereas in Python it can be done using a single line of code. The syntax used in Python is